

Q1.

a)

to compute a neuron network output need to calculate weight sum of input and bias

formula = $(a1.w1) + (a2.w2) + \text{bias}$

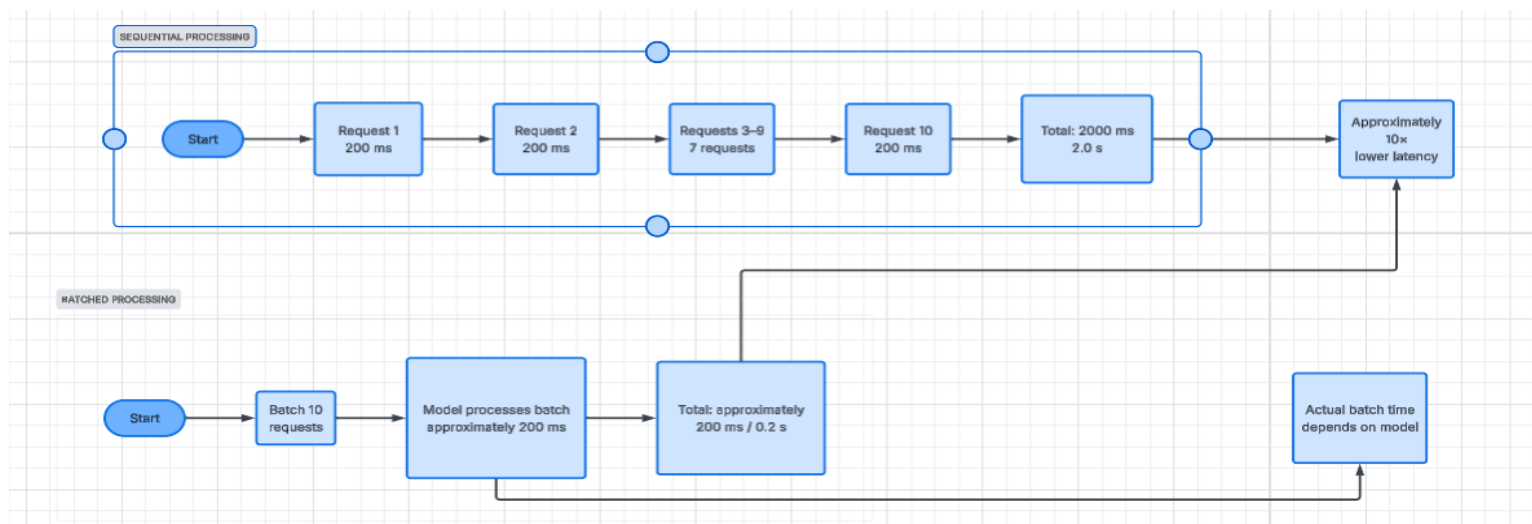
$x1=0$, $x2 = 3$, $w1 = 1.0$, $w2 = 0.2$, $\text{bias} = 0.5$

$= (0 * 1.0) + (3 * 0.2) + 0.5$

$= 0 + 0.6 + 0.5$

$= 1.1$

b)



c)

Activation function need to check wheater the the neuron input should pass furtur or not to output.
With the linear way no matter how many inputs it has it should give only single output.

Q2 Inference

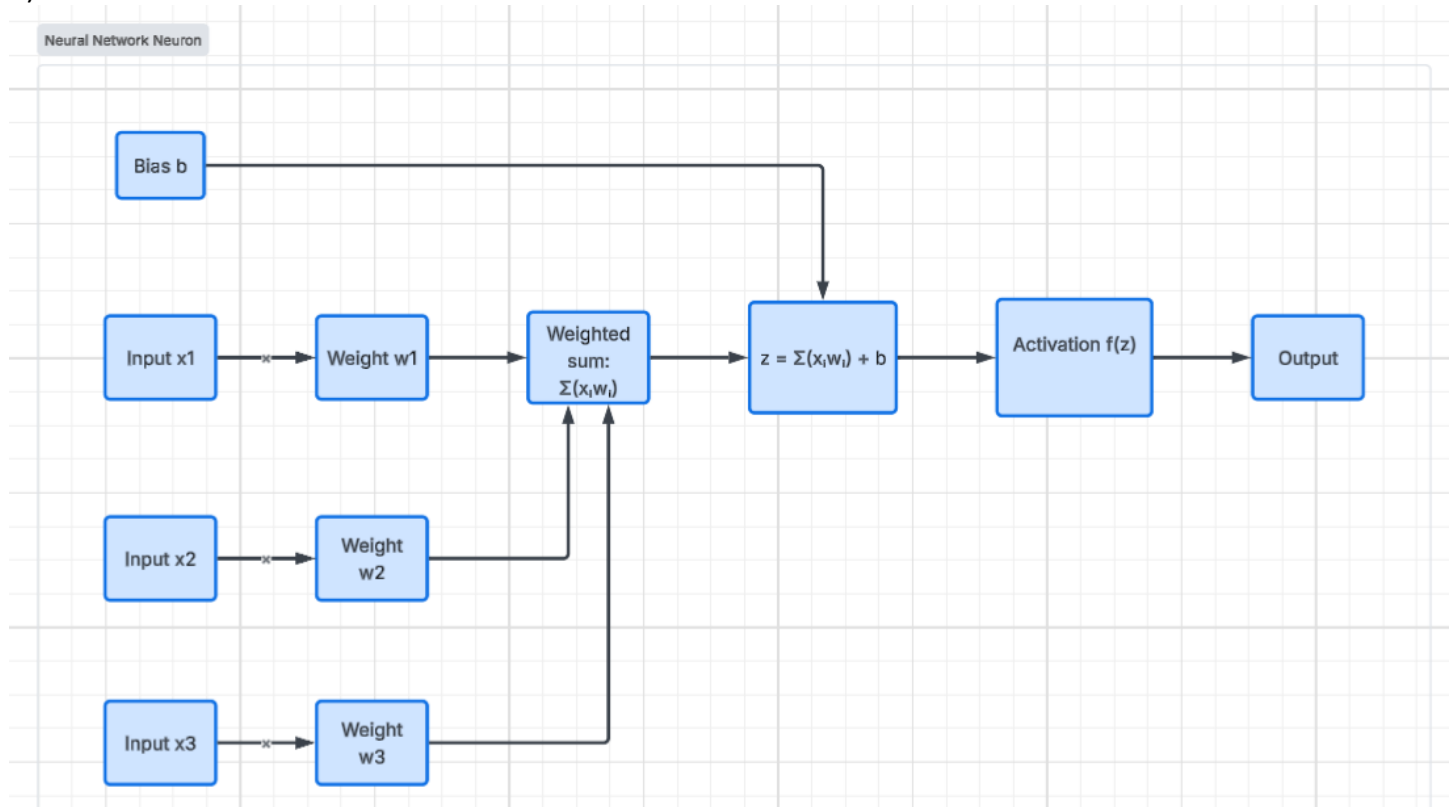
a)

if each request takes 200ms
and send 10 model request

total time = 200ms * 10

=2000ms

B)



C)

Why batch over sequential

1. Batch process save Reduce latency in between the requests.
2. Improve efficiency by performing multiple tasks.

3.LLMs

Q1

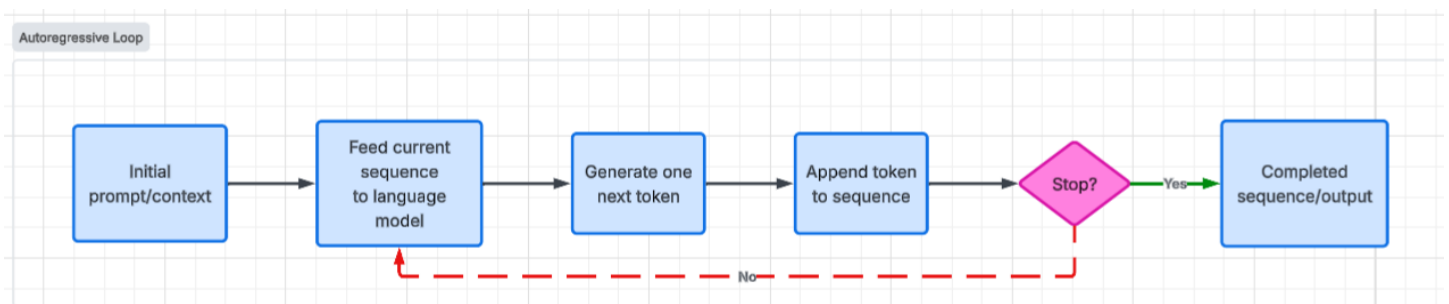
A)

Sentence = "LLMs generate text one token at a time"

words = 8

Tokens = 8

B)



C)

To generate a new word/token required entire previous sequence of tokens. As the size of output increases, the need to store the previous words to track them in KV-CACHE.

Each token is fully dependent on the previous text. That's why longer output takes more time to generate.